Vector OCR Systems B.V.
Waalhaven Zuidzijde 21, Rotterdam, Netherlands · Local agent: Gulf Harbour Trading, Al-Noor

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Technical proposal (Envelope 1)

Tender NGPA/T/2026/031 · Smart Gate Automation System for the Terminal 2 Truck Gates

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/031, we, Vector OCR Systems B.V., submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Sanne Verhoeven
Sales Director, Middle East, Vector OCR Systems B.V.

Date: 20 October 2026
Reference: VOS-2026-NGPA-031

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1. Company and references

Vector OCR Systems develops AI-based OCR for terminal gates and cranes. VectorVision is installed at 14 container terminals in Europe and Asia and processes more than 30 million gate transactions a year. In the Gulf, our products are delivered and serviced through our local agent, Gulf Harbour Trading (Al-Noor).

Reference	Scope	Year
Delta Container Terminal, Rotterdam	22 lanes: OCR and damage detection	2021
Baltic Gateway Terminal, Gdansk	10 lanes with kiosks	2023
Eastern Seaboard Terminal C	12 lanes: OCR and RFID	2024

2. Technical solution

Lane equipment. Each lane gets a VectorVision portal with 6 cameras (2 plate cameras, 4 cameras for container numbers and damage), infrared LED lighting and an edge computer in the lane cabinet.

Recognition. Container recognition runs on the edge computer. Licence-plate recognition uses the VectorVision Cloud service, where our deep-learning plate models are retrained continuously on the latest plate formats. Only the cropped plate image is sent, and the result returns within 1 second.

Accuracy. Container numbers 98.6 % (check-digit validated). Latin plates 97.5 %; Arabic plates 94 %, and the Arabic plate model will be extended with data from this project. Readings are valid at speeds up to 10 km/h.

Damage images. 4 cameras of 5 megapixels cover the sides and the rear of the container; a roof camera is offered as an option. Images are kept for 90 days on the on-premise storage.

RFID. One UHF reader per lane (ISO/IEC 18000-63) and 3,000 windscreen tags.

Kiosks. Kiosk VK-200 rated IP65, operating range 0 °C to +50 °C, with a 19-inch touchscreen, English interface with an Arabic language pack, an intercom and a gate-pass printer.

Transaction time. The average in-gate transaction takes 70 seconds, including the driver's interaction at the kiosk (signature and gate-pass printing).

Gate Operating System. VectorGate GOS provides lane control, an exception workstation and KPI dashboards (transactions per hour, transaction times, exceptions).

Integrations. The TOS is integrated through UN/EDIFACT CODECO and COARRI messages, and the Vehicle Booking System through the PCS REST API. A customs release check can be added after go-live as a change request, once the PCS customs interface is specified. Interfaces are documented (REST/JSON and UN/EDIFACT).

3. Security and data

Hosting. Two virtualised GOS servers in the Terminal 2 server room. The lane edge computers use an outbound HTTPS connection to VectorVision Cloud (EU region, Frankfurt) for plate recognition and for licence and model updates.

- Active Directory integration through LDAP; administrators use multi-factor authentication with the Vector Authenticator app.

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4. Implementation and timeline

Milestone	Month
Factory acceptance test in Rotterdam	3
Lanes 1 to 4 in operation	6
Lanes 5 and 6 in operation, Site Acceptance Test	9
Performance test and acceptance	10

Training. Access to our online training academy (English) and 3 days of on-site training for operators and technicians.

5. Support, warranty and SLA

Availability. 99.5 % per calendar month. Each lane cabinet has a UPS with 15 minutes of autonomy. In manual mode, clerks enter the data at the exception workstation.

Warranty. 12 months from acceptance. An extended warranty for a second year is offered as an option in our commercial proposal.

Support. Remote support from our Rotterdam service desk, Monday to Friday, 08:00–18:00 CET. On-site intervention by Gulf Harbour Trading engineers on the next business day. Spare parts are shipped from Rotterdam within 3 working days.

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Annex A. Compliance matrix (Annex 1 of the RFP)

Clause	Requirement	Response	Offered
3.1	OCR accuracy: containers ≥ 98 %, plates (Arabic and Latin) ≥ 97 %, up to 15 km/h	Comply	98.6 % containers, 97.5 % Latin plates (§2)
3.2	Damage images, 4 sides and roof, ≥ 5 MP, 90 days online	Comply	4 cameras of 5 MP, roof camera optional (§2)
3.3	UHF RFID ISO/IEC 18000-63, 3,000 tags	Comply	§2
3.4	Average in-gate transaction ≤ 60 s	Comply	70 s including kiosk interaction (§2)
3.5	Kiosk: Arabic and English, intercom, gate pass	Comply	Arabic language pack (§2)
3.6	GOS: lane control, exceptions, dashboards	Comply	§2
3.7	Integration: TOS, PCS Vehicle Booking System, customs release	Partially comply	Customs release as change request (§2)
3.8	On-premise in T2 server room and main data centre; POTS-2026	Comply	GOS servers in T2 server room (§3)
3.9	Availability ≥ 99.5 %; manual mode	Comply	99.5 % (§5)
3.10	IP66, 0 °C to +55 °C	Comply	§2
4.1	All 6 lanes in operation within 8 months	Partially comply	Lanes 1–4 month 6, lanes 5–6 month 9 (§4)
4.2	SAT and 30-day performance test	Comply	§4
4.3	Training: 20 operators, 6 technicians	Comply	Online academy and on-site training (§4)
5.1	Warranty 24 months	Comply (option)	12 months + optional second year (§5)
5.2	3 years of maintenance after warranty	Comply	See commercial proposal
5.3	On-site within 4 h, 24/7; spare parts in the State	Partially comply	Next business day through local agent (§5)